Laboratory 6 – Exercises

1. Use the following dataset:

```
X, y = make_blobs(
        n_samples=400,
        centers=4,
        cluster_std=1.25,
        random_state=42
       ),
```

in order to apply the original K-means clustering algorithm. Run the algorithm for 40 iterations and plot the model's decision boundaries.

2. The following function is used to create a dataset with 2 features:

```
def Circle(r,n):
    return [(math.cos(2*math.pi/n*x)*r+
            +np.random.normal(-30,30),math.sin(2*math.pi/n*x)*r+
            +np.random.normal(-30,30)) for x in range(1,n+1)]
```

This function takes the radius and number of data points as arguments and returns an array of data points which when plotted forms a circle. In order to apply clustering algorithms, three concentric circles are created, and finally noise is added to the data as follows:

```
df=pd.DataFrame(Circle(500,1000))
df=df.append(Circle(300,700))
df=df.append(Circle(100,300))

X=df.append([(np.random.randint(-600,600),
              np.random.randint(-600,600)) for i in range(300)])
```

Perform the following:

➢ plot the dataset;

➢ apply the K-means++ algorithm and plot the decision boundaries;

➢ apply the agglomerative clustering algorithm with complete linkage and visualize the clusters;

➢ apply the DBSCAN algorithm with `eps=35` and `min_samples=6`, and plot the clusters.